PRISTINI
SCHOOL OF AI

Chapter 1 Introduction

Dr. Ameni Youssfi Nouira

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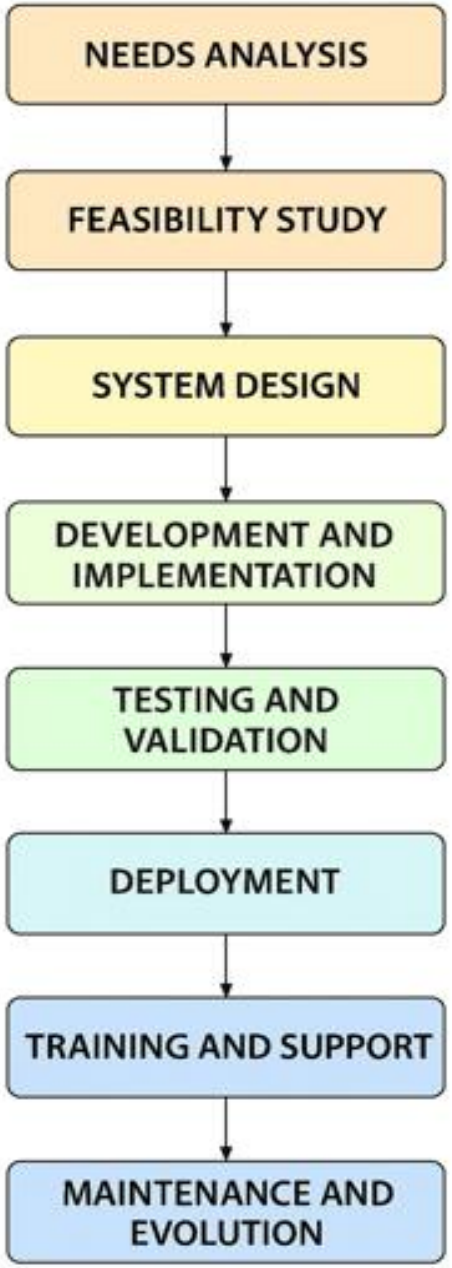
Outline

- Introduction To Information system
- Introduction to architecture
- UML
- Object Paradigm

IS

- An information system represents the set of elements involved in the management, storage, processing, transmission, and dissemination of information within an organization.
- Roles:
  - information collection,
  - information storage,
  - information processing,
  - information dissemination.

The main steps in the development of an information system are:



Steps in the development of an information system

- **Requirements expression:** definition of a requirements specification document.
- **Specification:** what the system should be and how it can be used.
- **Analysis:** elements involved in the information system, their structures and relationships. Three axes must be defined:
  - **Object know-how → Functional axis**
  - **Object structure → Static axis**
  - **Object life cycle → Dynamic axis**
- **Design:** providing technical solutions — architecture, performance, and optimization — as well as the definition of structures and algorithms.

Steps in the development of an information system

- **Implementation:** development and programming.
- **Testing and verification:** quality controls carried out throughout the development cycle.
- **Validation:** checking compliance with the requirements specification and discussion with the user.
- **Maintenance and evolution:**
  - **Corrective maintenance:** fixing bugs.
  - **Adaptive/evolutionary maintenance:** integrating new changes.

Design

- A creative process for developing software
- Provides an architecture for the software by dividing it into modules, each responsible for different functionalities
- **Two types of design**
- ✓ **Architectural design**
  - Definition of the internal structure of the software
  - Decomposition into manageable-sized components
  - Definition of interfaces and interactions between components
- ✓ **Detailed design**
  - Definition of the role of each component
  - Definition of subcomponents
- *(Object-Oriented Design Methodology)*

Modeling

- **Support for design**
- **Formalization of the solution**, using notations or known rules
- **Helps reduce the complexity of a phenomenon**
  - Eliminate non-significant details
  - Reflect what is important for understanding and predicting the modeled phenomenon
- **Creation of a Model**
  - An abstract and simplified representation of a real-world entity
  - Aimed at describing, explaining, or predicting it
  - *Example: Plan*

Methods and Languages

- **Design Method**
  - Normative description of the steps of modeling
  - Example: *Merise*, *UML*
  - **Problem:**
    - The existence of several specific cases ⇒ difficulty in representing an exhaustive method
- **Modeling Language**
  - A graphical language to represent and communicate the different aspects of an information system
  - Contains a vocabulary and rules that allow combining elements in order to communicate

UML

History

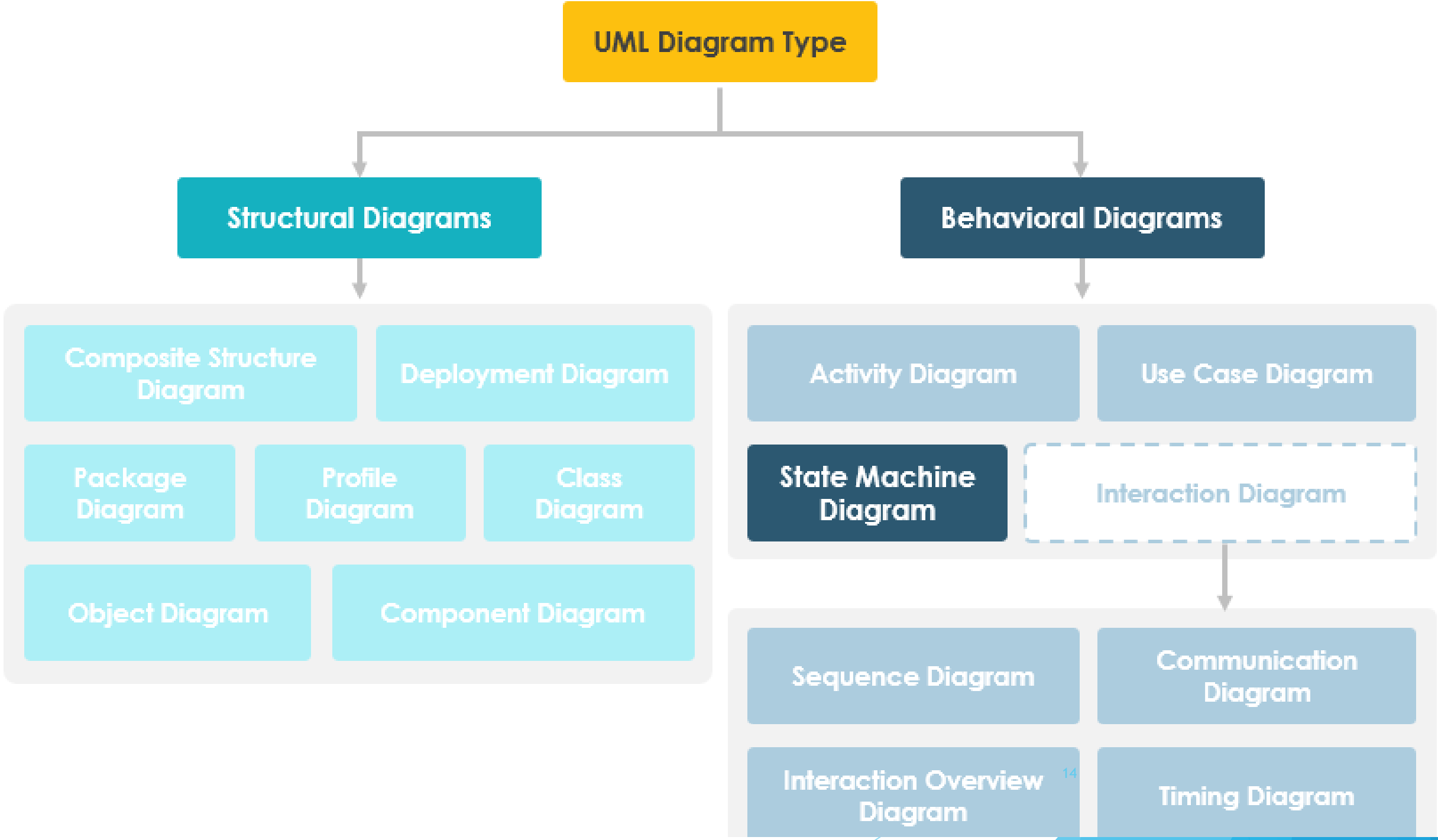
- **1980s:**
  ✓ Methods to organize functional programming (Merise)
  ✓ Separation of data and processes
- **Early 1990s:**
  ✓ Emergence of object-oriented programming: need for an adapted methodology
  ✓ Emergence of more than 50 methods between 1990 and 1995
- **1994:**
  ✓ Consensus on 3 methods:
  - **OMT** by James Rumbaugh: graphical representation of the static, dynamic, and functional aspects of a system
  - **OOD** by Grady Booch: package concept
  - **OOSE** by Ivar Jacobson: description of user requirements

UML

- Consensus among the "3 Amigos" to create a common method:
  ✓ **UML: Unified Modeling Language**
- **1997:** UML was defined as the standard for object-oriented information system modeling by the **OMG (Object Management Group)**
- UML is used across all areas of software development:
  ✓ Information systems
  ✓ Telecommunications
  ✓ Transport, aeronautics, aerospace
  ✓ Scientific fields
- But not only that: it can also be used to model mechanical, human, and other behaviors.

Basic Building Blocks of UML

- **Elements**
  ✓ These are the essential abstractions of the model.
- **Relationships**
  ✓ Relationships express the existing links between different elements.
- **Diagrams**
  ✓ A diagram is a visual representation of all the elements that make up the system.
  ✓ They help visualize a system from different perspectives (e.g., user, administrator).
  ✓ In complex systems, a single diagram only provides a partial view of the system:
  - All diagrams together provide an overall view of the system to be designed.
  - Each diagram helps model or specify a particular view (specific aspect) of the system to be designed.


UML Diagram Type
Structural Diagrams
Behavioral Diagrams
Composite Structure Diagram
Deployment Diagram
Package Diagram
Profile Diagram
Class Diagram
Object Diagram
Component Diagram
Activity Diagram
Use Case Diagram
State Machine Diagram
Interaction Diagram
Sequence Diagram
Communication Diagram
Interaction Overview Diagram
Timing Diagram